DANIEL KARIMI

TELECOMMUNICATIONS ENGINEER & AI

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ABOUT ME

Telecommunications Systems Engineer graduated from **Universidad Politécnica de Madrid**, with a strong background in communication systems, networking, and software development. My approach combines technical skills in **programming**, **data processing**, and **cloud & DevOps tools** with a problem-solving mindset oriented towards building reliable, scalable solutions.

I am passionate about **low-level Artificial Intelligence**: from training and developing my own LLM models to implementing software solutions that solve both technical and everyday problems. I firmly believe in understanding how things work from their foundations.

I am looking for opportunities where I can combine my knowledge in **telecommunications** with the development of **innovative AI and software solutions**, creating technology that has a real impact on the world.

TECHNICAL STACK

MACHINE LEARNING & AI

PyTorch / TensorFlow • Transformers / LLMs • NLP Concepts • YOLO / Computer Vision • Deep Learning • Distributed Training basics

PROGRAMMING & PIPELINES

Python • C / C++ • Data Pipelines • Data Curation • SQL / PostgreSQL • Pandas / NumPy

DEVELOPMENT

Python • JavaScript / TypeScript • C • React.js • Next.js • HTML5 / CSS3 • Node.js • SQL

CLOUD & DEVOPS

Vercel • Cloudflare Workers • GitHub Actions • CI/CD • Docker • Linux • Git

DATA SCIENCE

Pandas • NumPy • Data Curation • Data Analysis • Visualization • Jupyter

TOOLS

Git / GitHub • VS Code • PostgreSQL • REST API • Agile Methodologies • Documentation

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council)

PROFESSIONAL EXPERIENCE

AI Engineer Intern | CSIC – Institute of Physical and Information Technologies (ITEFI)

Oct 2025 – Jan 2026

- Developed automated **data processing pipelines** for biological image analysis: conversion, normalization, and optimization of large datasets.
- Designed, trained, and fine-tuned cutting-edge Computer Vision models using **PyTorch** and **YOLOv8**, ensuring high reliability for scientific research.
- Built and maintained **end-to-end training and deployment pipelines**, handling dataset generation, curation, and quality assurance.
- Performed rigorous **performance analysis and optimization** of ML models: evaluating metrics, adjusting hyperparameters, and improving predictive accuracy.
- Collaborated closely with researchers, translating novel complex requirements into elegant data structures and engineering solutions.
- Configured and managed distributed development environments (Anaconda/Python) and resolved dependency issues across multiple platforms.

PyTorch | Python | Data Pipelines | Model Training | Research

EDUCATION

Telecommunications Systems Engineering | **Universidad Politécnica de Madrid (ETSIT-UPM)** *Sep 2020 – Jul 2025*

- **Programming & Algorithms:** Data structures, OOP, algorithm design and optimization (C, Python, Java).
- **Networking & Systems:** TCP/IP architecture, network protocols, operating systems, distributed systems.
- **Digital Signal Processing:** Spectral analysis, digital filtering, real-time DSP algorithms.
- **Electronics & Embedded Systems:** Microprocessors, analog/digital circuits, hardware-software integration.

Java | C | Python | Networking | DSP | Algorithms

Technology Baccalaureate, **IES Isabel la Católica** *2018 – 2020*

Science and Technology track. Honours. Strong foundation in mathematics and physics.

CERTIFICATIONS

AI and Productivity (Google): Efficient use of AI tools and automation workflows. •

Generative AI (MIT): Implementation of GenAI solutions and sentiment analysis. •

Internet of Things (MIT): IoT solutions for smart homes and precision agriculture. •

Digital Transformation (MIT): Technology strategy and business digitalization. •

Aptis ESOL B2 (British Council): Official English certification accrediting full linguistic competence (CEFR). Assessment of Reading, Writing, Listening and Speaking, demonstrating ability to work in professional environments and understand complex technical language.

KEY PROJECTS

CellTracker AI + SCIENTIFIC PAPER

PYTHON • YOLOV8 • PYTORCH • OPENCV

AI-powered cell tracking system for microscopy. Pending publication in scientific journal.

Applied ML project for biology research. Implements custom PyTorch-based detection models for automatic cell identification and tracking. Includes an elegant end-to-end data pipeline: preprocessing, multi-object tracking (Kalman filters), quantitative trajectory analysis, and metrics export.

- Custom-trained PyTorch/YOLOv8 detection and multi-object tracking.
- End-to-end pipeline: preprocessing, trajectory analysis, and metrics export.

MigraCell Analyzer DEEP LEARNING + BIO

PYTHON • YOLOV8-SEG • PYTORCH • TKINTER

Intelligent Microscopy Platform for Cell Migration Analysis.

Full automation of wound healing assays using Deep Learning. Processes complete experiments in < 5 minutes (vs 4-8h manually). Inference with YOLOv8-seg, mask post-processing, geometric computation, and automatic report generation.

- 100% reproducible results with pixel-perfect YOLOv8-seg segmentation.
- Unattended batch processing and publication-ready chart generation.